

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Experiment No.: PBL Q.1 (CO-1)

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1. TITLE

Interactive Product Calculator using JavaScript Functions and Event Listeners

2. AIM

To create a webpage containing a product name, quantity and price, and to calculate the total price dynamically using JavaScript functions and event listeners whenever the quantity or price changes, without refreshing the page.

3. OBJECTIVE

The objectives of this experiment are:

- To create an HTML form with product name, quantity and price fields.
- To write reusable JavaScript functions.
- To attach event listeners to input elements.
- To read and validate user input using JavaScript.
- To update the DOM dynamically without reloading the page.

4. THEORY

- JavaScript is a scripting language that runs in the browser and makes web pages interactive.
- The Document Object Model (DOM) represents the page as objects. `document.getElementById()` is used to access elements.
- An event is an action such as typing or clicking. `addEventListener("input", function)` runs a function every time the value of an input field changes.
- A function groups statements into a reusable block. Here, `calculateTotal()` validates the inputs and computes the total.
- `parseFloat()` converts the text in an input field to a number; `isNaN()` and `Number.isInteger()` are used for validation.
- `textContent` is used to change the text of an element, so the result is displayed without refreshing the page.
- Formula used: Total Price = Quantity x Price.

The basic flow of the application is:

User Input -> input event -> calculateTotal() -> Validation -> Calculation -> DOM Update

5. ALGORITHM / PROCEDURE

1. Create an HTML page with input fields for product name, quantity and price.
2. Add elements to display an error message, a summary and the total price.
3. Access all elements in JavaScript using `document.getElementById()`.
4. Define the function `calculateTotal()`.

5. Inside the function, read the quantity and price and convert them to numbers.
6. Validate the quantity (whole number, minimum 1) and the price (0 or more).
7. If the input is invalid, display an error message and show a total of 0.00.
8. Otherwise calculate total = quantity x price.
9. Update the summary and total price in the page using `textContent`.
10. Attach an "input" event listener to the product, quantity and price fields.
11. Call `calculateTotal()` once when the page loads to show the initial state.
12. Open the page in a browser and test with multiple valid and invalid inputs.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Interactive Product Calculator</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #eef2f7;
      display: flex;
      justify-content: center;
      padding: 40px 10px;
    }
    .card {
      background: #ffffff;
      width: 380px;
      padding: 24px 28px;
      border-radius: 10px;
      box-shadow: 0 2px 10px rgba(0, 0, 0, 0.15);
    }
    h2 { text-align: center; color: #1f3a5f; margin-top: 0; }
    label { display: block; margin-top: 14px; font-weight: bold; }
    input {
      width: 100%;
      padding: 8px;
      margin-top: 5px;
      box-sizing: border-box;
      font-size: 15px;
    }
    .error {
      color: #c62828;
      min-height: 20px;
      margin-top: 12px;
      font-size: 14px;
    }
    .result {
      margin-top: 10px;
      padding: 14px;
      background: #e8f5e9;
      border-left: 5px solid #2e7d32;
      border-radius: 4px;
    }
    .result .summary { font-size: 14px; color: #444; min-height: 18px; }
```

```

        .result .total { font-size: 24px; font-weight: bold; color: #1b5e20; }
    </style>
</head>
<body>
    <div class="card">
        <h2>Product Calculator</h2>

        <label for="productName">Product Name</label>
        <input type="text" id="productName" value="Notebook">

        <label for="quantity">Quantity</label>
        <input type="number" id="quantity" value="1" min="1" step="1">

        <label for="price">Price per Unit (Rs.)</label>
        <input type="number" id="price" value="0" min="0" step="0.01">

        <div class="error" id="errorMessage"></div>

        <div class="result">
            <div class="summary" id="summary"></div>
            <div class="total" id="totalPrice">Total: Rs. 0.00</div>
        </div>
    </div>

    <script>
        // Get references to the HTML elements
        const productInput = document.getElementById("productName");
        const quantityInput = document.getElementById("quantity");
        const priceInput = document.getElementById("price");
        const errorMessage = document.getElementById("errorMessage");
        const summary = document.getElementById("summary");
        const totalPrice = document.getElementById("totalPrice");

        // Function to validate the inputs and calculate the total price
        function calculateTotal() {
            const productName = productInput.value.trim() || "Product";
            const quantity = parseFloat(quantityInput.value);
            const price = parseFloat(priceInput.value);

            errorMessage.textContent = "";

            // Validation
            if (isNaN(quantity) || quantity < 1 || !Number.isInteger(quantity)) {
                errorMessage.textContent =
                    "Quantity must be a whole number (minimum 1).";
                summary.textContent = "";
                totalPrice.textContent = "Total: Rs. 0.00";
                return;
            }
            if (isNaN(price) || price < 0) {
                errorMessage.textContent =
                    "Price must be a valid number (0 or more).";
                summary.textContent = "";
                totalPrice.textContent = "Total: Rs. 0.00";
                return;
            }
        }

        // Calculation and DOM update (no page refresh)
        const total = quantity * price;
    </script>

```

```

        summary.textContent = quantity + " x " + productName +
            " @ Rs. " + price.toFixed(2);
        totalPrice.textContent = "Total: Rs. " + total.toFixed(2);
    }

    // Event listeners: recalculate whenever any input changes
    productInput.addEventListener("input", calculateTotal);
    quantityInput.addEventListener("input", calculateTotal);
    priceInput.addEventListener("input", calculateTotal);

    // Show the initial state when the page loads
    calculateTotal();
</script>
</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<code>document.getElementById()</code>	Selects an HTML element using its id.
<code>function calculateTotal()</code>	Reads the inputs, validates them, calculates the total and updates the page.
<code>parseFloat(value)</code>	Converts the text entered in the field into a number.
<code>isNaN(), Number.isInteger()</code>	Validate that the quantity is a whole number and the price is a valid number.
<code>quantity * price</code>	Calculates the total price.
<code>textContent</code>	Updates the text of an element without reloading the page.
<code>toFixed(2)</code>	Displays the amount with two decimal places.
<code>addEventListener("input", ...)</code>	Calls <code>calculateTotal()</code> every time the user changes a field.
<code>calculateTotal();</code>	Shows the initial result when the page loads.

8. TEST CASES AND EXECUTION DATA

The test cases were executed in a browser and the actual results were recorded.

Test Case	Product	Qty	Price	Expected Result	Actual Result	Status
TC-01	Pen	10	25	Total: Rs. 250.00	Total: Rs. 250.00	Pass
TC-02	Notebook	3	45.50	Total: Rs. 136.50	Total: Rs. 136.50	Pass
TC-03	Laptop Bag	2	1299.99	Total: Rs. 2599.98	Total: Rs. 2599.98	Pass
TC-04	Pen	10 -> 12	25	Total: Rs. 300.00 (updates without reload)	Total: Rs. 300.00 (page not reloaded)	Pass
TC-05	Mouse	0	350	Error: Quantity must be a whole number (minimum 1).	Error: Quantity must be a whole number (minimum 1).	Pass
TC-06	Mouse	2	-5	Error: Price must be a valid number (0 or more).	Error: Price must be a valid number (0 or more).	Pass

9. OUTPUT

Output 1: Initial page (Quantity = 1, Price = 0)

Product Calculator

Product Name

Quantity

Price per Unit (Rs.)

1 x Notebook @ Rs. 0.00

Total: Rs. 0.00

Output 2: TC-01 - Pen, Quantity 10, Price 25

Product Calculator

Product Name

Quantity

Price per Unit (Rs.)

10 x Pen @ Rs. 25.00

Total: Rs. 250.00

Output 3: TC-04 - Quantity changed to 12; total updated instantly

Product Calculator

Product Name

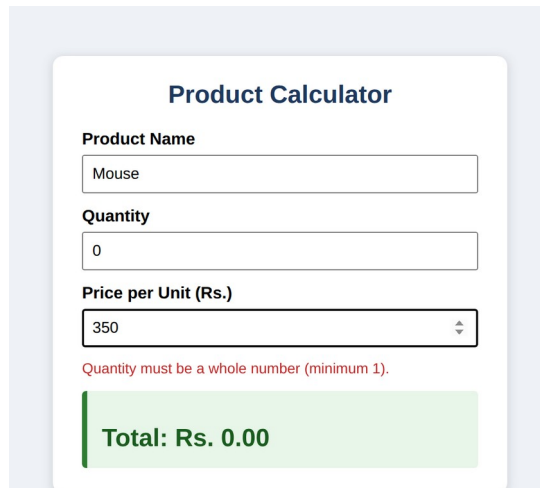
Quantity

Price per Unit (Rs.)

12 x Pen @ Rs. 25.00

Total: Rs. 300.00

Output 4: TC-05 - Invalid quantity (0) shows an error message



The screenshot displays a web form titled "Product Calculator". It contains three input fields: "Product Name" with the value "Mouse", "Quantity" with the value "0", and "Price per Unit (Rs.)" with the value "350". Below the "Quantity" field, a red error message states: "Quantity must be a whole number (minimum 1)". At the bottom of the form, a green box displays the calculated total: "Total: Rs. 0.00".

10. RESULT

Thus, an interactive product calculator was successfully developed using HTML and JavaScript. The total price is calculated dynamically using functions and event listeners whenever the product details change, without refreshing the page, and the application was verified using multiple test cases.